User Document

This project suggests an approach to solve numerically Burgers' Equation in turbulent and laminar flow mode (different value of diffusion coefficient). This project represents small part of my research area which focuses on the numerical study of turbulent flows.

1. Mathematics and Theory

1.1. Burgers' Equation

Burgers' equation is a fundamental partial differential equation occurring in various areas of applied mathematics, such as fluid mechanics, nonlinear acoustics, gas dynamics, traffic flow. For a given field $u(x,t)$ and diffusion coefficient (or viscosity, as in the original fluid mechanical context) ϑ, the general form of Burgers' equation (also known as viscous Burgers' equation) in one space dimension is the following dissipative system:

$$\frac{du}{dt} + u\frac{du}{dx} = \vartheta\frac{d^2u}{dx^2} \; ; \; 0 \leq x \leq 1 \; and \; t \geq 0 \qquad (1)$$

This project presents numerical solution of Burgers' equation for laminar (low Reynolds number) and turbulent modes. Turbulent mode happens in high Reynold number which is equivalent to low viscosity.

1.2 Finite Difference Method

Finite difference method (FDM) is suggested as the numerical technique to solve this partial differential equation. The FDM approximation of time derivative, first and second derivatives of field $u(x,t)$ can respectively be expressed as:

$$\frac{du}{dt} = \frac{(u(x,t+dt)-u(x,t))}{dt} \qquad (2)$$

$$\frac{du}{dx} = \frac{(u(x+dx,t)-u(x-dx,t))}{2\,dx} \tag{3}$$

$$\frac{d^2u}{dx^2} = \frac{(u(x+dx,t)+u(x-dx,t)-2\,u(x,t))}{(dx)^2} \tag{4}$$

The proposed approach can solve the Burgers' equation via DNS (direct numerical solution) and LES (large eddy simulation).

1.3. Direct Numerical Solution

A direct numerical simulation (DNS) is a simulation in computational fluid dynamics in which the Burgers' equation, and in general Naveir-Stokes equation, are numerically solved without any turbulence model. Therefore, high resolution is demanded to capture the smallest details involved in a physical domain.

1.4. Large Eddy Simulation

Large eddy simulation (LES) is a mathematical model for turbulence used in computational fluid dynamics. The principal idea behind LES is to reduce the computational cost by ignoring the smallest length scales, which are the most computationally expensive to resolve. In this project, Smagorinsky model which introduces an artificial viscosity to our solution used as a LES model. Therefore, the total viscosity enhances and LES viscosity stabilizes the numerical calculation for lower resolution.

$$S = \frac{d}{dx}(\vartheta_e \frac{du}{dx}) \tag{5}$$

$$\vartheta_e = (C_s\delta)^2 \left|\frac{du}{dx}\right| \tag{6}$$

Since S is similar to diffusion term in Burgers' equation, right hand side of Eq (1), ϑ_e represents the mentioned artificial viscosity and it is added to normal viscosity. Thus, term S is added to the right hand side of Eq (1).

2. Input File

The code provides an option for users testing their own input files as the initial condition ($u(x); t = 0$). Making a sample input file is easy for this software package:

1. Open a text file and then write title "`Variable u`" for your field $u(x)$ at any line.

2. Copy the value of given field $u(x_0)$ for point x_0 in the next line. Then, repeat this instruction for $u(x_1)$ in the following line. Please repeat this action until you insert the data of your initial function for all grids.
   Notice: this package only accepts structured grid points (with the same grid size for all grids). Also, the values of $u(x_i)$ should be written in order based on an increasing order of x_i:
   $x_0 < x_1 < \cdots < x_i < x_{i+1} < \cdots < x_n$; Grid resolution=n+1 (7)

3. Change the name of input file to "`example.txt`" and copy that in the folder that `Makefile` Were located. ("`Solver`")
   Some input files as examples were included in "`Examples`" folder. You may add more information to your input file as it can be observed in the example files. However, it is not necessary.

3. Solver Application

When users run the executable file, some questions appear on the screen. The software asks the users to set input file, the value of viscosity and final time of solution and the type of solution (DNS or LES). It also shows the brief information of your simulation at the end of program.

If you choose LES option, when the program completes, some output files are generated with the name "`outputLES.txt`". Each file includes the time of solution, grid position and calculated field $u(x,t)$.

The DNS option produces two types of output files, "`outputDNS.txt`" and "`outputDNSFiltered.txt`". "`outputDNS.txt`" contains the DNS solution and the second file include the filtered results from DNS solution. The second file can be used to compare your DNS solution with LES data.

4. Postprocessing

The data can be plotted in Excel software. In the input file, the grid positons and the calculated field $u(x,t)$ were located under title "`Variable 'x'`" and "`Variable 'u'`" respectively. Also, the time of solution mentioned at the beginning of each output file.